AI-Enhanced ESG Reporting: Technical Research Brief

Multi-Framework Report Generation from Unstructured Company Data

Research Question

How can large language models reliably generate consistent, framework-aligned ESG reports from identical input data across multiple reporting standards?

Technical Challenge

The Multi-Framework Problem

ESG reporting requires compliance with multiple frameworks:

- **IFRS S1/S2** (regulatory standard, climate-focused)
- **GRI Standards** (voluntary, multi-stakeholder)
- **VSME** (SME-proportionate, EU standard)

The challenge: Same underlying company data → completely different output requirements

Aspect	IFRS S1/S2	GRI Standards	VSME
Focus	Climate risk + sustainability disclosures	Environmental, Social, Governance	Proportionate SME reporting
Structure	Regulatory sections	GRI Standards (101-400 series)	Simplified elements
Vocabulary	Climate/financial terminology	Stakeholder engagement terminology	SME-accessible language
Length	4,000 tokens	5,000 tokens	4,000 tokens
Overlap with IFRS	100% (baseline)	22.7%	~15%

Problem: Can't simply "rewrite" one report for another framework—fundamentally different requirements demand different content organization.

Contact & Collaboration

Researcher: Dr Huamao Wang | Associate Professor in Finance, Risk and Banking **Institution:** University of Nottingham **Email:** huamao.wang@nottingham.ac.uk **Research Topics:** LLM evaluation, compliance automation, multi-output generation

Interested in collaboration on:

- Methodology validation or critique
- Comparative model study
- Commercial prototype development
- Further research